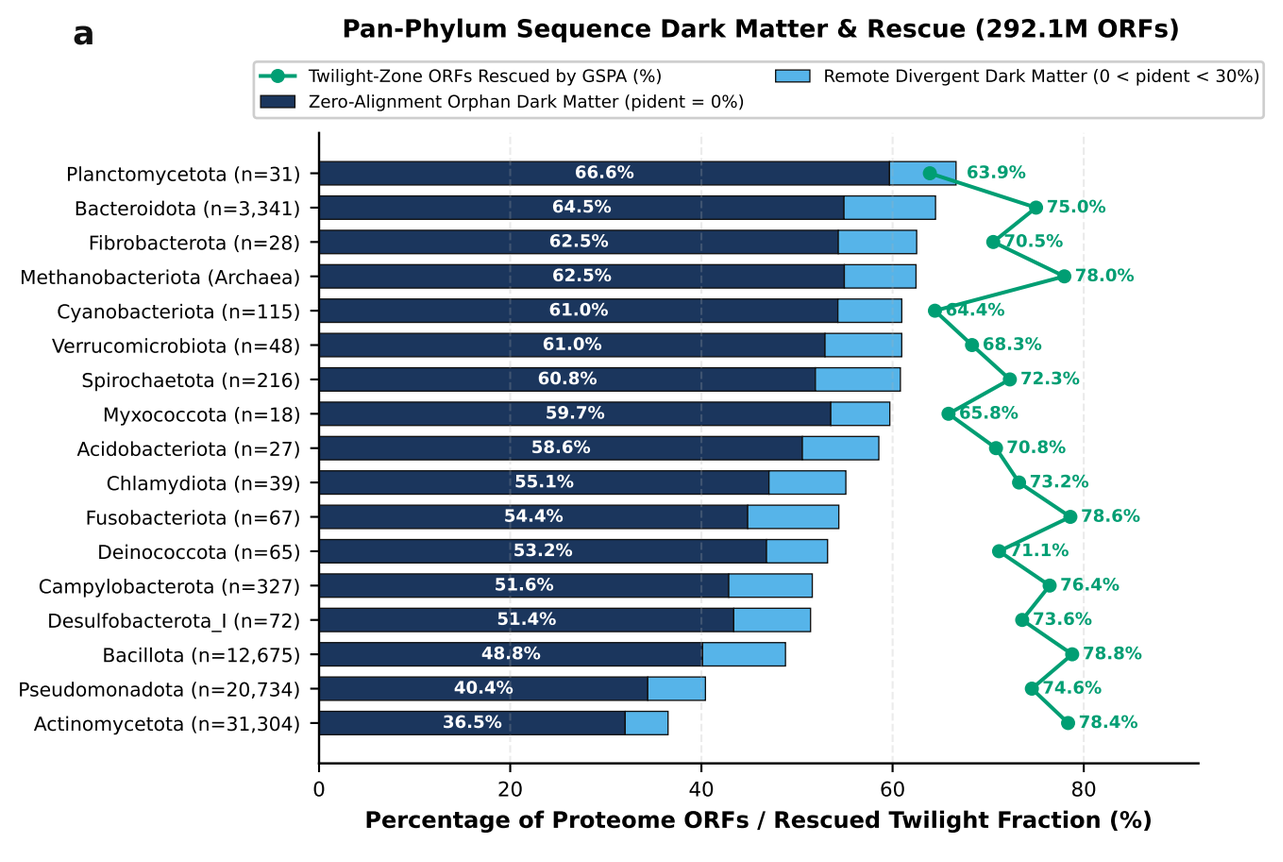
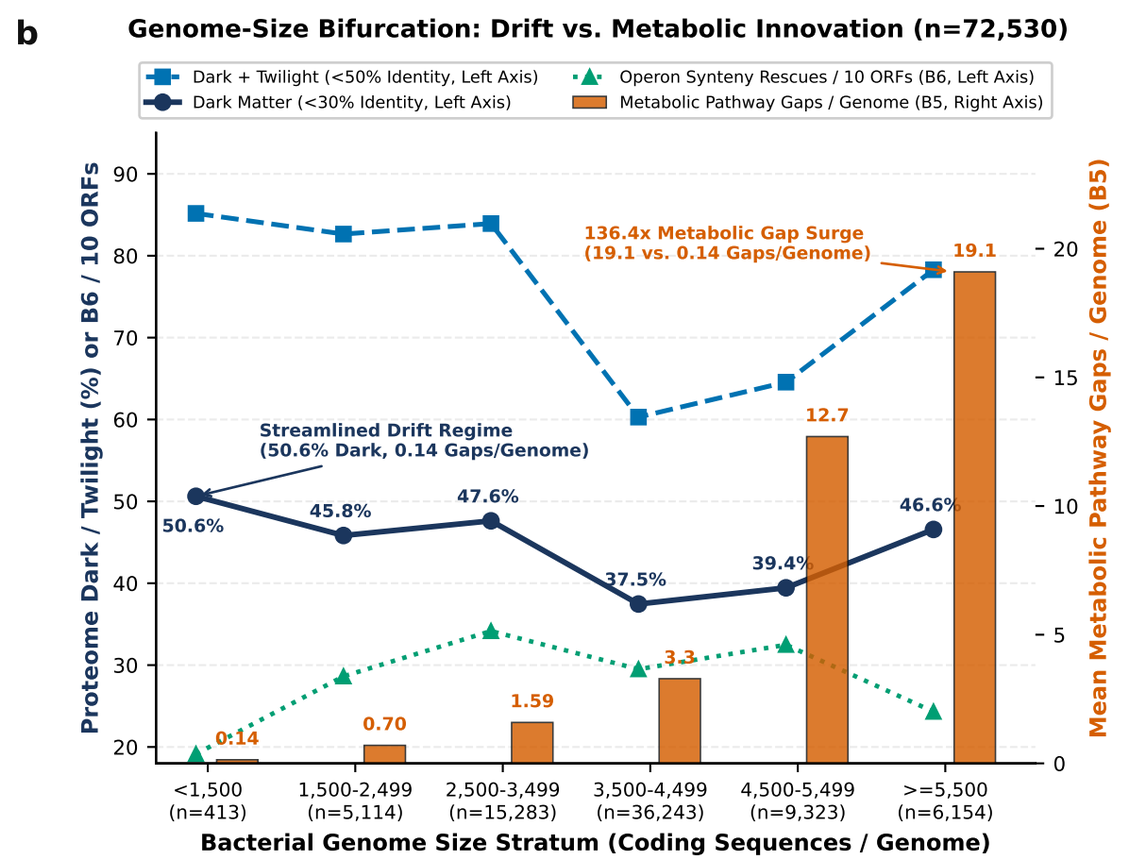
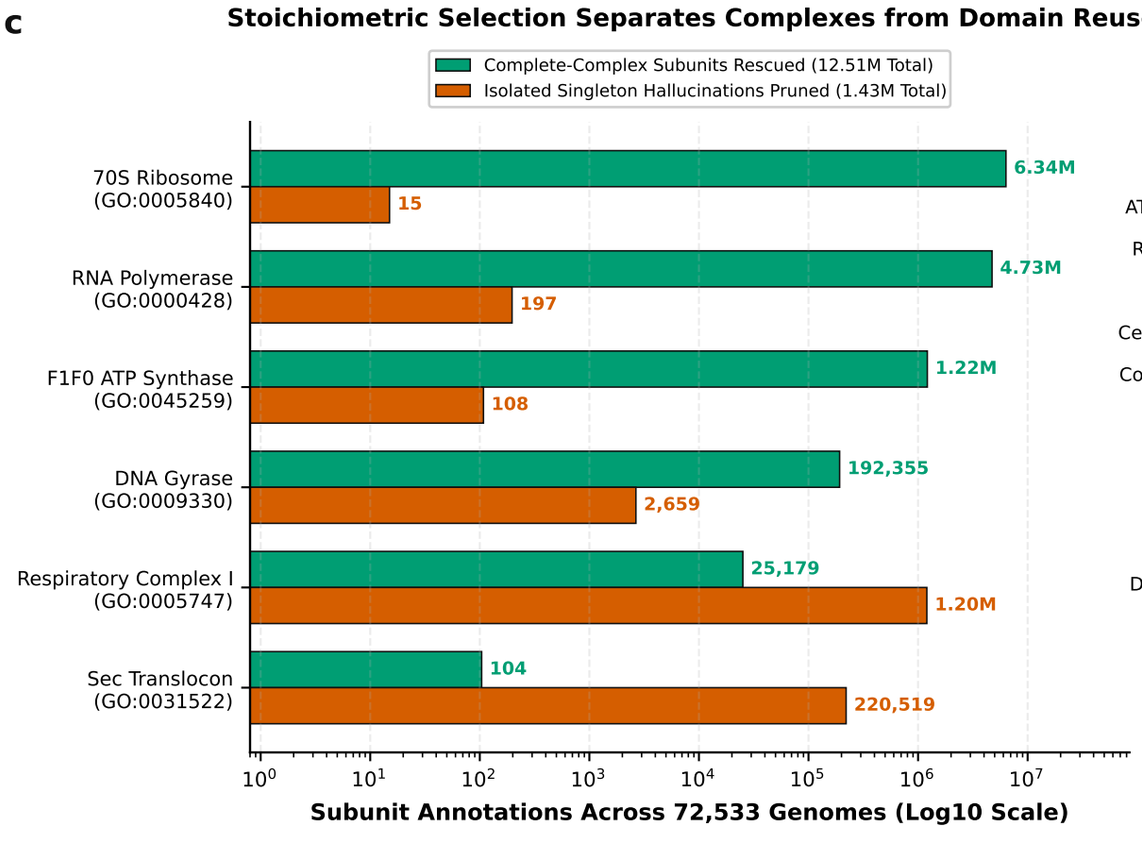
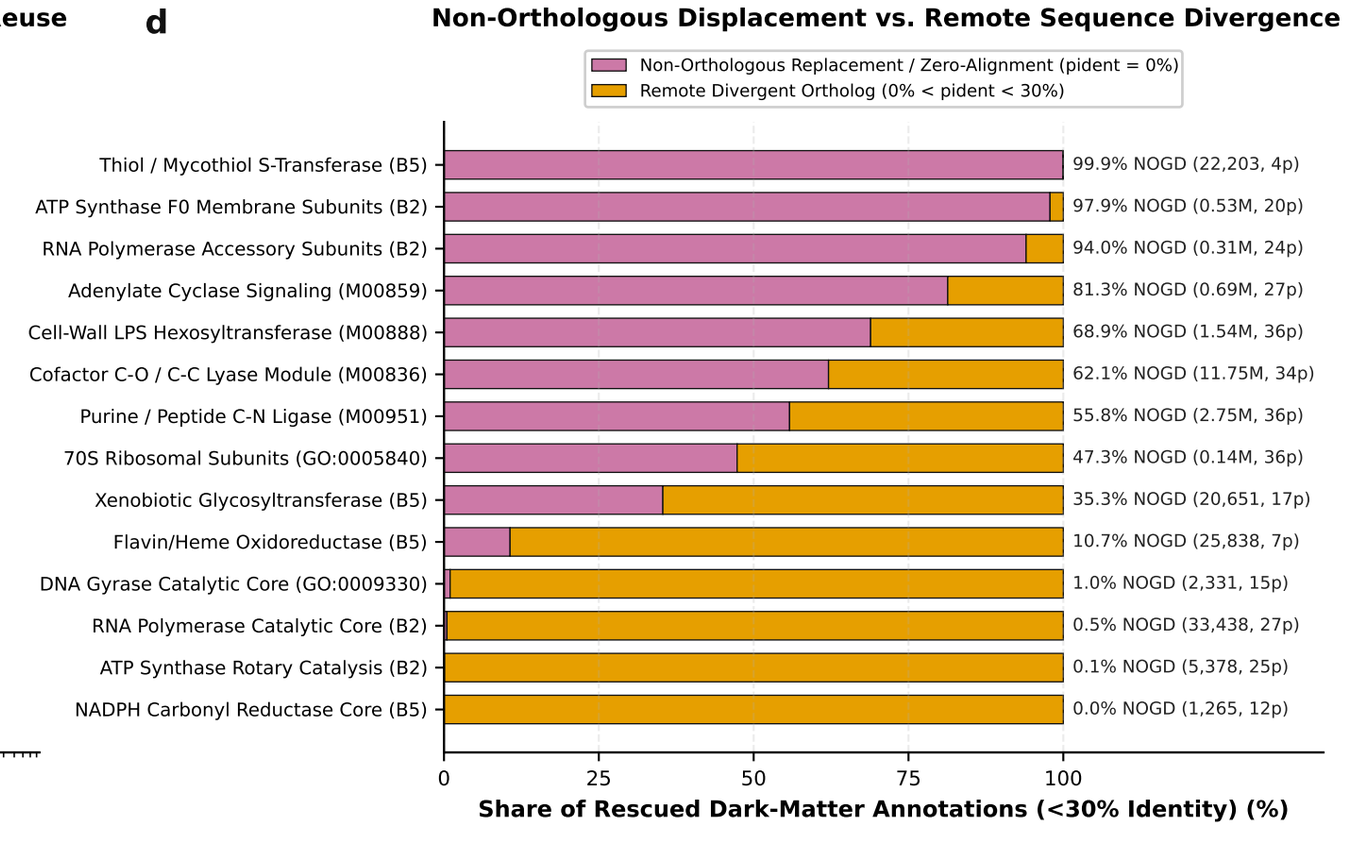

a
Pan-Phylum Sequence Dark Matter & Rescue (292.1M ORFs)
Twilight-Zone ORFs Rescued by GSPA (%)
Remote Divergent Dark Matter (0 < pident < 30%)
Zero-Alignment Orphan Dark Matter (pident = 0%)
Planctomycetota (n=31)
Bacteroidota (n=3,341)
Fibrobacterota (n=28)
Methanobacteriota (Archaea)
Cyanobacteriota (n=115)
Verrucomicrobiota (n=48)
Spirochaetota (n=216)
Myxococcota (n=18)
Acidobacteriota (n=27)
Chlamydiota (n=39)
Fusobacteriota (n=67)
Deinococcota (n=65)
Campylobacterota (n=327)
Desulfobacterota_I (n=72)
Bacillota (n=12,675)
Pseudomonadota (n=20,734)
Actinomycetota (n=31,304)
66.6%
64.5%
62.5%
62.5%
61.0%
61.0%
60.8%
59.7%
58.6%
55.1%
54.4%
53.2%
51.6%
51.4%
48.8%
40.4%
36.5%
63.9%
75.0%
70.5%
78.0%
64.4%
68.3%
72.3%
65.8%
70.8%
73.2%
78.6%
71.1%
76.4%
73.6%
78.8%
74.6%
78.4%
0
20
40
60
80
Percentage of Proteome ORFs / Rescued Twilight Fraction (%)


b
Genome-Size Bifurcation: Drift vs. Metabolic Innovation (n=72,530)
Dark + Twilight (<50% Identity, Left Axis)
Dark Matter (<30% Identity, Left Axis)
Operon Synteny Rescues / 10 ORFs (B6, Left Axis)
Metabolic Pathway Gaps / Genome (B5, Right Axis)
Proteome Dark / Twilight (%) or B6 / 10 ORFs
Mean Metabolic Pathway Gaps / Genome (B5)
136.4x Metabolic Gap Surge
(19.1 vs. 0.14 Gaps/Genome)
Streamlined Drift Regime
(50.6% Dark, 0.14 Gaps/Genome)
50.6%
45.8%
47.6%
37.5%
39.4%
46.6%
0.14
0.70
1.59
3.3
12.7
19.1
<1,500
(n=413)
1,500-2,499
(n=5,114)
2,500-3,499
(n=15,283)
3,500-4,499
(n=36,243)
4,500-5,499
(n=9,323)
>=5,500
(n=6,154)
Bacterial Genome Size Stratum (Coding Sequences / Genome)


c
Stoichiometric Selection Separates Complexes from Domain Reuse
Complete-Complex Subunits Rescued (12.51M Total)
Isolated Singleton Hallucinations Pruned (1.43M Total)
70S Ribosome
(GO:0005840)
RNA Polymerase
(GO:0000428)
F1F0 ATP Synthase
(GO:0045259)
DNA Gyrase
(GO:0009330)
Respiratory Complex I
(GO:0005747)
Sec Translocon
(GO:0031522)
6.34M
15
4.73M
197
1.22M
108
192,355
2,659
25,179
1.20M
104
220,519
Subunit Annotations Across 72,533 Genomes (Log10 Scale)


d
Non-Orthologous Displacement vs. Remote Sequence Divergence
Non-Orthologous Replacement / Zero-Alignment (pident = 0%)
Remote Divergent Ortholog (0% < pident < 30%)
Thiol / Mycothiol S-Transferase (B5)
ATP Synthase F0 Membrane Subunits (B2)
RNA Polymerase Accessory Subunits (B2)
Adenylate Cyclase Signaling (M00859)
Cell-Wall LPS Hexosyltransferase (M00888)
Cofactor C-O / C-C Lyase Module (M00836)
Purine / Peptide C-N Ligase (M00951)
70S Ribosomal Subunits (GO:0005840)
Xenobiotic Glycosyltransferase (B5)
Flavin/Heme Oxidoreductase (B5)
DNA Gyrase Catalytic Core (GO:0009330)
RNA Polymerase Catalytic Core (B2)
ATP Synthase Rotary Catalysis (B2)
NADPH Carbonyl Reductase Core (B5)
99.9% NOGD (22,203, 4p)
97.9% NOGD (0.53M, 20p)
94.0% NOGD (0.31M, 24p)
81.3% NOGD (0.69M, 27p)
68.9% NOGD (1.54M, 36p)
62.1% NOGD (11.75M, 34p)
55.8% NOGD (2.75M, 36p)
47.3% NOGD (0.14M, 36p)
35.3% NOGD (20,651, 17p)
10.7% NOGD (25,838, 7p)
1.0% NOGD (2,331, 15p)
0.5% NOGD (33,438, 27p)
0.1% NOGD (5,378, 25p)
0.0% NOGD (1,265, 12p)
Share of Rescued Dark-Matter Annotations (<30% Identity) (%)